Conditioning AttentionDCA on PCA coordinates, and adding fields: experiments C1, C3 and M2

AttentionDCA / J-domains — follow-up to the proposal note

October 2, 2026

Abstract

Three of the proposed methods are implemented and tested on the clean split: **C1**, a harmonic restraint on the PCA coordinates added to the trained model at sampling time; **C3**, a model with fields that depend on the target coordinates, learned by conditional pseudo-likelihood; and **M2**, plain per-site fields added to AttentionDCA. §1 and §2 are a slow refresher on the two ideas everything rests on: how an energy function turns into sampling probabilities (and what entropy and free energy have to do with cluster weights), and why a PCA coordinate is an exact Potts term. The results follow in §4–§6.

1 A statistical-physics refresher

Energy and probability. A Potts model assigns every sequence $\boldsymbol{x} = (x_1, \dots, x_L)$ an energy $E(\boldsymbol{x})$ and defines its probability by the Boltzmann distribution

$$p(\boldsymbol{x}) = \frac{e^{-\beta E(\boldsymbol{x})}}{\mathcal{Z}}, \qquad \mathcal{Z} = \sum_{\boldsymbol{x}} e^{-\beta E(\boldsymbol{x})}, \tag{1}$$

with $\beta = 1/T$ the inverse temperature. Our model has $E(\boldsymbol{x}) = -\sum_i h_i(x_i) - \frac{1}{2}\sum_{i\neq j} J_{ij}(x_i, x_j)$ (the main report's model has $h = 0$). The partition function $\mathcal{Z}$ is a sum over 21^{63} sequences and is never computed; it is not needed, because everything we do uses *ratios* of probabilities, in which it cancels:

$$\frac{p(\boldsymbol{x})}{p(\boldsymbol{y})} = e^{-\beta\,[E(\boldsymbol{x}) - E(\boldsymbol{y})]}. \tag{2}$$

At $\beta = 1$, a sequence one energy unit lower is $e \approx 2.7$ times more probable, ten units lower $e^{10} \approx 22\,000$ times. This is why energies are quoted in *nats*: an energy difference *is* a log-probability ratio.

How the sampler uses the energy. One step of the sampler picks a site i, holds the other $L-1$ letters fixed, and asks: what would the energy be for each of the 21 possible letters at i? It then draws letter a with probability

$$p(x_i = a \mid \boldsymbol{x}_{-i}) = \frac{e^{-\beta E(\boldsymbol{x} \text{ with } x_i = a)}}{\sum_b e^{-\beta E(\boldsymbol{x} \text{ with } x_i = b)}} = \mathrm{softmax}_a\left[\beta\Big(h_i(a) + \sum_{j\neq i} J_{ij}(a, x_j)\Big)\right]. \tag{3}$$

Only the terms touching site i survive (the rest is common to all 21 options and cancels), which is why one update is cheap. Low-energy letters are favoured but not chosen deterministically, and β sets how strongly: as $\beta \to \infty$ the update always takes the best letter (a greedy descent that freezes in a single minimum), as $\beta \to 0$ it picks letters uniformly at random. Repeating (3) over all sites is one *sweep*. A chain started from any sequence forgets it after a few autocorrelation times, and from then on visits each sequence a fraction of the time equal to its Boltzmann probability (1) — this is detailed balance, and it is exact when $J_{ij}(a, b) = J_{ji}(b, a)$. (Attention couplings are slightly asymmetric, so for us it holds approximately; the main report discusses this.)

Why chains do not sit at the minimum: entropy. The energy along a chain relaxes from the starting value to a *typical* value and then fluctuates around it; it does not keep going down to the lowest-energy sequence. The reason is counting. Let $\Omega(E)$ be the number of sequences with energy near E and $S(E) = \ln \Omega(E)$ the **entropy**. The probability of finding the chain at energy E is (number of such sequences) $\times$ (probability of each):

$$P(E) \propto \Omega(E)\, e^{-\beta E} = e^{\,S(E) - \beta E}. \tag{4}$$

Low energies are individually likely but rare; medium energies are individually less likely but astronomically more numerous. The chain lives where $S(E) - \beta E$ is largest. Lowering the temperature (raising β) shifts the balance towards energy, which is exactly what $\beta^* > 1$ did in the main report: it pushed the samples towards lower energy and stronger correlations.

From sequences to regions: free energy. Now ask for the probability of a whole region k of sequence space, for instance a PCA cluster:

$$P(k) = \sum_{\boldsymbol{x}\in k} p(\boldsymbol{x}) = \frac{1}{\mathcal{Z}} \sum_{\boldsymbol{x}\in k} e^{-\beta E(\boldsymbol{x})} \equiv \frac{e^{-\beta F_k}}{\mathcal{Z}}, \qquad F_k = -\frac{1}{\beta} \ln \sum_{\boldsymbol{x}\in k} e^{-\beta E(\boldsymbol{x})}. \tag{5}$$

F_k is the **free energy** of the region. It is an exact identity (take the restricted distribution $p_k(\boldsymbol{x}) = e^{-\beta E(\boldsymbol{x})}/e^{-\beta F_k}$ on k and compute its Shannon entropy $S_k = -\sum_{\boldsymbol{x}\in k} p_k \ln p_k = \beta\langle E\rangle_k - \beta F_k$) that

$$F_k = \langle E\rangle_k - T\, S_k, \tag{6}$$

the mean energy of the region minus temperature times its entropy. A region carries a large weight if its sequences have low energy *or* if there are very many of them. Mean energy alone does not decide.

What s_k is. In the main report s_k denotes a cluster's **share**: the fraction of sequences (probability mass) that falls in cluster k, i.e. $s_k = P(k)$ above — not the entropy, which we write S_k. The two are related through (5)–(6): the share is $s_k \propto e^{-\beta F_k}$, and the entropy enters through F_k. The free-energy error of the model for a cluster is

$$\Delta F_k = -\ln \frac{s_k^{\text{gen}}}{s_k^{\text{nat}}} \quad \text{nats},$$

the amount by which the model's F_k is too high (relative to the other clusters) to reproduce the natural share. $\Delta F = 1$ means the cluster gets $1/e \approx 0.37$ of its natural share.

The entropy is visible in our data. Applying (6) to the samples of the baseline model gives, relative to the dominant cluster C1, $\langle E\rangle_k - \langle E\rangle_{C1} = (22.5,\ 0.0,\ 9.3)$, $F_k - F_{C1} = (2.8,\ 0.0,\ 2.2)$ and therefore $S_k - S_{C1} = (19.7,\ 0.0,\ 7.1)$ for $k =$ C0, C1, C2. The under-sampled cluster C0 sits about 22.5 energy units above C1 but only 2.8 nats above it in free energy: the difference, 19.7 nats, is entropy. C0 is a wide, diverse basin (its sequences are also the most mutually distant, §5). If energy alone set the weights, C0 would get a share $e^{-22.5}$ of C1's; entropy is what keeps it at a few percent.

How a restraint changes the picture. Adding a term $U(\boldsymbol{x})$ to the energy multiplies every probability by $e^{-\beta U(\boldsymbol{x})}$ and renormalises. With $U = \frac{\lambda}{2}\|\mathrm{PC}(\boldsymbol{x}) - \boldsymbol{t}\|^2$ the factor is a Gaussian window of width $1/\sqrt{\beta\lambda}$ around the target $\boldsymbol{t}$: sequences far from $\boldsymbol{t}$ are suppressed, sequences near it keep their relative Boltzmann weights. That is precisely a conditional distribution, which §2 makes explicit.

2 The PCA argument, step by step

Step 1: a sequence as a vector of zeros and ones. PCA works on numbers, so each sequence is first one-hot encoded: for every site i and letter a there is an entry $x_{ia} \in \{0, 1\}$, equal to 1 iff site i carries letter a. A sequence becomes a vector of length $Lq = 63 \times 21 = 1323$ with exactly one 1 per site. Toy example with $L = 3$ and a two-letter alphabet $\{\mathrm{A}, \mathrm{G}\}$:

$$\mathrm{AGA} \;\mapsto\; (\underbrace{1,0}_{\text{site 1}} \mid \underbrace{0,1}_{\text{site 2}} \mid \underbrace{1,0}_{\text{site 3}}).$$

Step 2: a PCA coordinate is a dot product. Fitting PCA finds directions $\boldsymbol{v}_c$ (unit vectors of length 1323) along which the natural sequences vary most. Fitting is a nonlinear procedure, but once it is done, the coordinate of a sequence is just its projection on $\boldsymbol{v}_c$:

$$\mathrm{PC}_c(\boldsymbol{x}) = \boldsymbol{v}_c \cdot (\boldsymbol{x} - \boldsymbol{\mu}) = \sum_{i,a} v_{c,ia}\, x_{ia} - m_c, \qquad m_c = \boldsymbol{v}_c \cdot \boldsymbol{\mu}, \tag{7}$$

with $\boldsymbol{\mu}$ the mean one-hot vector. This is linear in $\boldsymbol{x}$; nothing is approximated.

Step 3: because $\boldsymbol{x}$ is one-hot, the dot product is a table lookup per site. At each site exactly one x_{ia} is 1, so the sum over a picks one number per site:

$$\mathrm{PC}_c(\boldsymbol{x}) = \sum_{i=1}^{L} V_c(i, x_i) - m_c, \qquad V_c(i, a) \equiv v_{c,ia}. \tag{8}$$

In the toy example with $\boldsymbol{v} = (0.5, -0.5\,|\,0.1, 0.3\,|\,-0.2, 0.4)$, PC(AGA) $= 0.5 + 0.3 - 0.2 - m$, and mutating site 3 from A to G moves the coordinate by $0.4 - (-0.2) = +0.6$, whatever the other sites are. The coordinate is a *score table*, like a position-specific scoring matrix: every site adds a contribution depending only on its own letter. For the J-domains a single substitution moves PC_1 by 0.125 on average, while the family spans about 6.9 units of PC_1 (standard deviation 2.06); going from one cluster to another takes tens of coordinated substitutions.

Step 4: the square of a sum of site terms is a Potts energy. A Potts energy is built from site terms $(h_i(x_i))$ and pair terms $(J_{ij}(x_i, x_j))$. Write $u_i = V_c(i, x_i)$ and expand

$$\Big(\sum_i u_i - m_c - t_c\Big)^2 = \underbrace{\sum_{i \neq j} u_i u_j}_{\text{pairs}} + \underbrace{\sum_i u_i^2 - 2(m_c + t_c)\sum_i u_i}_{\text{sites}} + \text{const.}$$

So the restraint $\frac{\lambda}{2}\sum_c (\mathrm{PC}_c(\boldsymbol{x}) - t_c)^2$ is exactly a Potts energy with

$$\Delta J_{ij}(a, b) = -\lambda \sum_c V_c(i, a) V_c(j, b), \qquad \Delta h_i(a) = \lambda \sum_c \big[(m_c + t_c) V_c(i, a) - \tfrac{1}{2} V_c(i, a)^2\big]. \tag{9}$$

Adding it to the learned energy gives *another Potts model*. The sampler is unchanged except that each site update also scores, for each candidate letter, how far the resulting coordinate would be from the target — a lookup in (8). Nothing is learned. (Unit test: on a toy system the sampler reproduces the enumerated distribution $\propto e^{-E - \lambda \|\mathrm{PC} - \boldsymbol{t}\|^2/2}$, and (9) matches the direct restraint to 10^{-10}.)

Step 5: what λ means — a noisy measurement. By Bayes' rule, if we "measure" the coordinate of a sequence with Gaussian noise of width σ and obtain $\boldsymbol{t}$, the posterior over sequences is

$$p(\boldsymbol{x} \mid \boldsymbol{t}) \;\propto\; p(\boldsymbol{x})\, e^{-\|\mathrm{PC}(\boldsymbol{x}) - \boldsymbol{t}\|^2 / 2\sigma^2}, \tag{10}$$

which is the restrained Boltzmann distribution with $\lambda = 1/\sigma^2$ (at $\beta = 1$). So λ is not a fudge factor: $1/\sqrt{\lambda}$ is the width of the target region, and $\lambda \to \infty$ is the exact conditional $p(\boldsymbol{x} \mid \mathrm{PC}(\boldsymbol{x}) = \boldsymbol{t})$. The old appended-label model tried to *learn* this relationship from data; here it is written down.

Step 6: why it cannot blow up the protein in the old way. The restraint only involves the two directions $\boldsymbol{v}_1, \boldsymbol{v}_2$. Everything else about the sequence — which residues are conserved, which pairs co-vary, the HPD motif — is still scored by the learned $E(\boldsymbol{x})$ alone. Raising λ narrows the window; it never amplifies the learned couplings. The price of a large λ is different: the chain must satisfy the coordinate almost exactly and may accept worse energies to do so. That trade-off is what the experiments measure.

Step 7: what C3 learns, in this language. C3 is trained on pairs (sequence, its coordinate plus noise of width σ), so the conditional it is fitted to is exactly (10) with $\lambda = 1/\sigma^2$. Expanding as in (9), that distribution has (i) a coupling correction $-\lambda \sum_c V_c V_c$ that does *not* depend on $\boldsymbol{t}$, and (ii) a field that is *linear* in $\boldsymbol{t}$. C3's parametrisation — shared couplings plus fields $h_i(a \mid \boldsymbol{z}) = \sum_f W_f(i, a)\phi_f(\boldsymbol{z})$ with ϕ containing $1, z_1, z_2$ — can therefore represent it: the learned J absorbs part (i), the linear features carry part (ii). We train C3 at $\sigma = 1,\ 0.5,\ 0.25$ and compare it with C1 at the matching $\lambda = 1, 4, 16$. This also predicts what happens if the $\boldsymbol{z}$-dependent field is amplified by a gain $\kappa > 1$ at sampling time, which is the analogue of the old β_{PCA}: the field part (ii) grows but the coupling part (i), which acts as the spring holding PC *at* the target, does not. Sequences are then pushed *past* the target — the "shifted too far right" of the previous study. §6 tests this prediction.

3 Setup

All models are trained on the clean split with the main report's architecture and hyper-parameters ($H = 128$, $d = 10$, $\lambda_J = 10^{-3}$, same seed and batches), and early stopping with a patience of 150 epochs; the only differences between runs are the field terms. Fields carry an L_2 penalty $10^{-4}\|W\|^2$.

Baseline field-free AttentionDCA (best epoch 962, validation NLPL 81.97).

M2 + per-site fields $h_i(a)$, learned jointly (82.03).

C1 the baseline plus the restraint (9) at sampling time; nothing is retrained. $\lambda \in \{0.25, 1, 4, 16, 64\}$.

C3 fields $h_i(a \mid \boldsymbol{z}) = \sum_f W_f(i,a)\phi_f(\boldsymbol{u})$, $\phi = (1, u_1, u_2, u_1^2, u_2^2, u_1u_2)$, $\boldsymbol{u}$ the standardised coordinates, trained on $(\boldsymbol{x}, \mathrm{PC}(\boldsymbol{x}) + \sigma\boldsymbol{\epsilon})$ with fresh noise at every batch, for $\sigma = 1,\ 0.5,\ 0.25$ (validation NLPL 81.54, 81.06, 80.02, scored with the noiseless coordinate). At sampling time $\boldsymbol{z} = \boldsymbol{t}$; a gain κ optionally multiplies the $\boldsymbol{z}$-dependent part of the field.

Targets are the three k-means centres of the main report (C0 = the under-sampled HscB-like cluster, C1, C2). For each method, target and setting, 3000 chains start from random *training* sequences — mostly in another cluster, so they have to travel — and run 500 sweeps. The restrained chains converge within 50 sweeps for $\lambda \geq 4$ and ~ 100 for $\lambda = 1$ (figure in the appendix). The PCA frame is the one used in every figure of the main report (fitted on the test set); it is a fixed linear map, V and $\boldsymbol{m}$ are just constants.

Metrics, all against the natural *test* sequences of the target cluster (640, 4873, 1888 sequences for C0, C1, C2; reweighted): distance $\|\mathrm{PC} - \boldsymbol{t}\|$; *overshoot*, the signed miss along the line from the family centre to the target (> 0: beyond the target, < 0: stopped short); share landing in the target cluster; $r(f_i)$ and $r(C_{ij})$; HPD motif retained (columns 28–30, present in $\geq 99\%$ of natural sequences); mean energy under one common judge (the baseline model, so C1 and C3 are on the same scale; natural cluster means -120.1, -135.2, -133.3); mean distance to the closest training sequence (d_{train}, novelty) and mean pairwise distance within the pool (div., diversity). Because C_{ij} estimates are noisy, *every* pool is evaluated at the same size, 3000; the natural ceiling is a weight-proportional draw of 3000 training sequences of the cluster. With these sizes the ceiling is $r(C_{ij}) = 0.659,\ 0.625,\ 0.750$ — lower than the full-family values of the main report simply because a cluster has fewer sequences.

4 M2: explicit fields

	val. NLPL	test NLPL	$r(f_i)$	$r(C_{ij})$ / slope	shares C0/C1/C2 (%)	ΔF_{C0}
natural	–	–	–	–	14/39/47	–
no fields	81.97	78.57	0.930	0.744 / 0.28	5/86/9	1.04
M2 (+ fields)	82.03	78.49	0.938	0.791 / 0.34	5/83/12	1.12

Table 1: Fields versus no fields, same training protocol. NLPL in nats per sequence; r and slope of the generated statistics against the natural test set (full family); shares by nearest cluster centre; $\Delta F_{C0} = -\ln(s^{\mathrm{gen}}_{C0}/s^{\mathrm{nat}}_{C0})$.

Fields are learned, but they replace couplings rather than add information. The fields are far from zero (mean $|h_i(a)| = 0.95$) and the coupling norm drops from 88.3 to 85.3, yet the pseudo-likelihood is unchanged (Table 1: validation 81.97 vs 82.03, test 78.57 vs 78.49). This confirms the premise of AttentionDCA: the conserved-residue and gap columns of the value matrices V^h already act as fields. The generated statistics improve modestly ($r(f_i)$ 0.930→0.938, $r(C_{ij})$ 0.744→0.791), so fields are a cheap, harmless addition.

Fields do not fix the missing cluster. C0 receives 5% of the samples with and without fields ($\Delta F_{C0} = 1.04 \to 1.12$). Figure 1 (right) shows why, using the decomposition $F = \langle E \rangle - S$ of §1. In both models C0 lies about 20 energy units above C1 and 17–20 nats higher in entropy; the two nearly cancel and leave a free-energy difference of 2.8–2.9 nats. A field shifts the energy of every sequence carrying a given residue; it cannot, by itself, rebalance two basins whose weights are set by a near-cancellation between energy and entropy. That balance is not what pseudo-likelihood measures — it fits each site's conditional given the rest of the sequence, never the relative mass of whole basins.

A finding worth more than M2 itself: the mode weights depend on when training stops. The main report's model stopped at epoch 668 (patience 50); retraining it with the current code reproduces its couplings bit for bit. The same run continued to epoch 962 (patience 150) has a *better* validation pseudo-likelihood (82.18→81.97, 0.2 nats per sequence) and the same C0 deficit, but its C1/C2 balance has flipped: shares C0/C1/C2 go from 5/36/59 to 5/86/9 (natural 14/39/47). That is a 2.7-nat swing in the relative free energy of C1 and C2 for 0.2 nats of pseudo-likelihood — and it makes the longer-trained model a *worse* generator of the family ($r(C_{ij}) = 0.744$, slope 0.28, against 0.87 for the report's model), because full-family correlations are dominated by the contrast between clusters. Two consequences: (i) the main report's statement that C1 and C2 are "covered" is true of that checkpoint but is not a robust property of the method; the C0 deficit, by contrast, is present in every model trained; (ii) the validation pseudo-likelihood cannot select a model for its sampling behaviour. Fixing the mode weights needs an objective that sees them (M4, likelihood/PCD fine-tuning) or, as below, an explicit choice of where to sample.

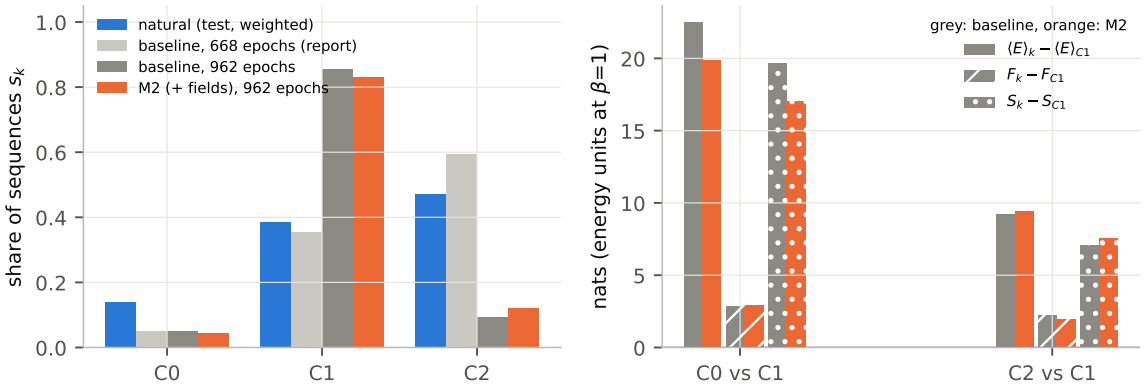


Figure 1: Left: cluster shares. The two grey bars (light, dark) are the *same* field-free model at two training lengths. Right: per-cluster mean energy, free energy and entropy of each model's own samples, relative to C1 ($F_k = -\ln s_k$, $S_k = \langle E\rangle_k - F_k$, $\beta = 1$).

5 C1: the analytic restraint

It steers, to any target, including the one the model neglects. Without restraint, 5%, 86% and 10% of the chains end in C0, C1, C2. From $\lambda = 4$ on, 100% end in the cluster they are aimed at (Figure 2), and the mean distance to the target falls as $1/\sqrt{\lambda}$ — 0.5, 0.3, 0.15 at $\lambda = 4, 16, 64$, the width of the Gaussian window of §2, Step 5. The response is monotone in λ, unlike β_{PCA} in the previous study.

It keeps the protein. The HPD motif is retained in 96–99% of the samples at every λ (Tables 2–4), against 99–100% for the natural clusters; $r(f_i)$ against the target cluster's natural sequences is 0.96–0.99 for $\lambda \geq 1$. The covariation agreement $r(C_{ij})$ *peaks* at a moderate stiffness — 0.458 for C0 and 0.605 for C2 at $\lambda = 4$, 0.49 for C1 at $\lambda = 0.25$ — against ceilings 0.659, 0.625, 0.750, and declines for larger λ. Most of that decline is not damage but a mismatch of scale: a natural cluster has a radius of about one PC unit, so a window of width 0.15 samples the *centre* of the cluster rather than the cluster, and the pool loses the cluster's internal variation (diversity drops from 0.56 to 0.47 for C1). **The useful regime is** $\lambda \approx 1$–4, a window comparable to the size of the region one wants.

Against rejection, the exact conditional. Keeping only the unconditional samples that fall in C0 gives 496 sequences out of 10 000; at the same cost C1 produces 10 000. Where the two can be compared they agree on what the model believes: C1 at $\lambda = 4$ and rejection give C0 sequences of the same energy (-106.3 vs -106.0), as they should if C1 samples the model's own conditional. That energy is 14 units above the natural C0 sequences (-120.1): the model's C0 basin is not only under-weighted (§4) but also displaced towards higher energy, and C1, which only restrains the coordinates, inherits it faithfully.

It inherits the prior's error at weak restraint. The overshoot column is negative at small λ: chains stop short of the target, on the side of the family centre (-1.42 for C0 at $\lambda = 1$, where only 80% reach the cluster). The restraint has to overcome the model's own free-energy penalty on C0, so the weaker it is, the more the model's mis-weighting shows through.

6 C3: the learned conditional field

At its natural setting it matches C1. Knowing the coordinate improves the conditional pseudo-likelihood by 0.4, 0.9 and 2.0 nats for $\sigma = 1$, 0.5, 0.25: the sharper the coordinate, the more it says about each site. Sampled at $\boldsymbol{z} = \boldsymbol{t}$ with $\kappa = 1$, C3 lands on target with the accuracy of C1 at the matching $\lambda = 1/\sigma^2$ (dashed vs solid curves of Figure 3; distances 0.42, 0.53, 0.49 at $\sigma = 0.5$ against 0.56, 0.54, 0.49 for C1 at $\lambda = 4$) and with comparable quality (HPD 99–100%, $r(C_{ij})$ within 0.07 of C1 for every target). This is the prediction of §2, Step 7: C3, trained on noisy coordinates, learns the same distribution that C1 writes down.

Where it is better: the neglected cluster. The one systematic difference is C0. C3 gets closer to it at weak conditioning ($\sigma = 1$: distance 0.74, overshoot -0.56, against 1.59 and -1.42 for C1 at $\lambda = 1$), and its C0 sequences have lower energy (-118.4 at $\sigma = 0.5$ against -106.3), closer to the natural C0 (-120.1). The reason is the prior. C1 multiplies the *unconditional* model, with its C0 free-energy error, by a window; C3 was trained

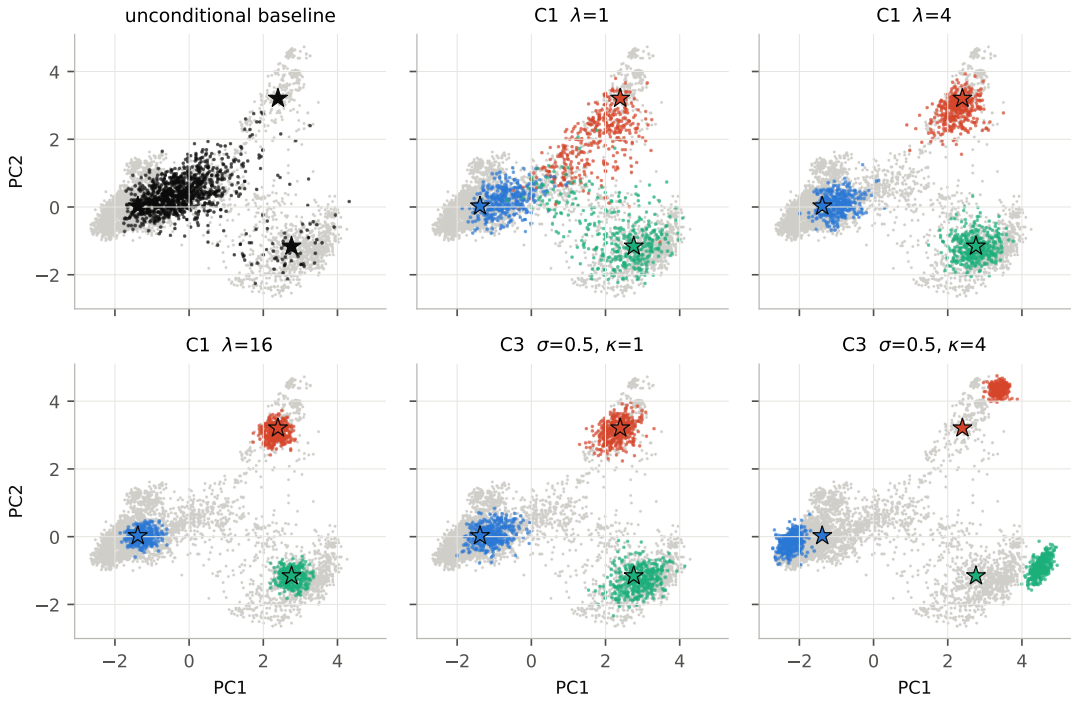


Figure 2: Samples in the PCA plane (grey: natural test). Stars: targets C0 (red), C1 (blue), C2 (green); points are coloured by the target they were aimed at. Top left: the unconditional baseline, which visits C1 almost exclusively. Bottom right: C3 with its target field amplified by $\kappa = 4$ — every pool lands *beyond* its target.

on (sequence, coordinate) pairs and learned the conditional directly from data, so it does not inherit that error. This is the one place where a learned conditional earns its extra parameters for a sequence-derived label.

Amplifying the target field reproduces the old failure, exactly as predicted. With the gain κ on the $\boldsymbol{z}$-dependent field the samples *overshoot*: +0.84, +1.36, +1.47 for C0 and +0.28, +0.84, +1.17 for C1 at $\kappa = 2, 4, 8$ (Figure 2, bottom right). Everything else degrades with it: diversity collapses to 0.16 (C0) and 0.21 (C1) — near-copies of one sequence — $r(C_{ij})$ falls to 0.08–0.12, and for C2 the HPD motif is lost in 44% of the samples while the energy rises to −81, 50 units above the natural cluster. The mechanism is the one derived in Step 7: κ amplifies the field that pushes towards the target but not the coupling term that holds the coordinate *at* the target. This is the β_{PCA} pathology of the previous study — "shifts too far right, loses the HPD motif" — now reproduced on demand and explained.

7 Summary

- **C1 works and needs no training.** Monotone control by λ, all three clusters reachable (C0 from 5% to 100%), HPD kept, best covariation agreement at $\lambda \approx 1$–4. It samples the model's own conditional, so it inherits the model's errors about a region (C0 is reached, but at too high an energy).
- **C3 matches C1 at** $\lambda = 1/\sigma^2$ and is better on the neglected cluster because it learns the conditional from data instead of reweighting a biased prior. Use it at $\kappa = 1$; amplifying it reproduces the old over-steering pathology.
- **M2** leaves the pseudo-likelihood unchanged, slightly improves the generated statistics, and does not fix the mode weights.
- **Mode weights are fragile under pseudo-likelihood**: 294 more epochs moved the C1/C2 balance by 2.7 nats for a 0.2-nat gain in validation PL, while C0 stayed at 5% in every model.

Next step suggested by these results. Conditioning gives a direct, cheap fix for the mode weights: draw a target $\boldsymbol{t}$ from the natural distribution of coordinates (the reweighted training PCA points), then sample $\boldsymbol{x} \sim p(\boldsymbol{x} \mid \boldsymbol{t})$ with C3 at $\kappa = 1$ (or C1 at $\lambda \approx 4$). This is ancestral sampling of $p(\boldsymbol{t})\, p(\boldsymbol{x} \mid \boldsymbol{t})$: the cluster shares are the natural ones *by construction*, and the within-cluster statistics are those measured here. It would remove both the C0 deficit and the training-length dependence of the C1/C2 balance without retraining anything but C3.

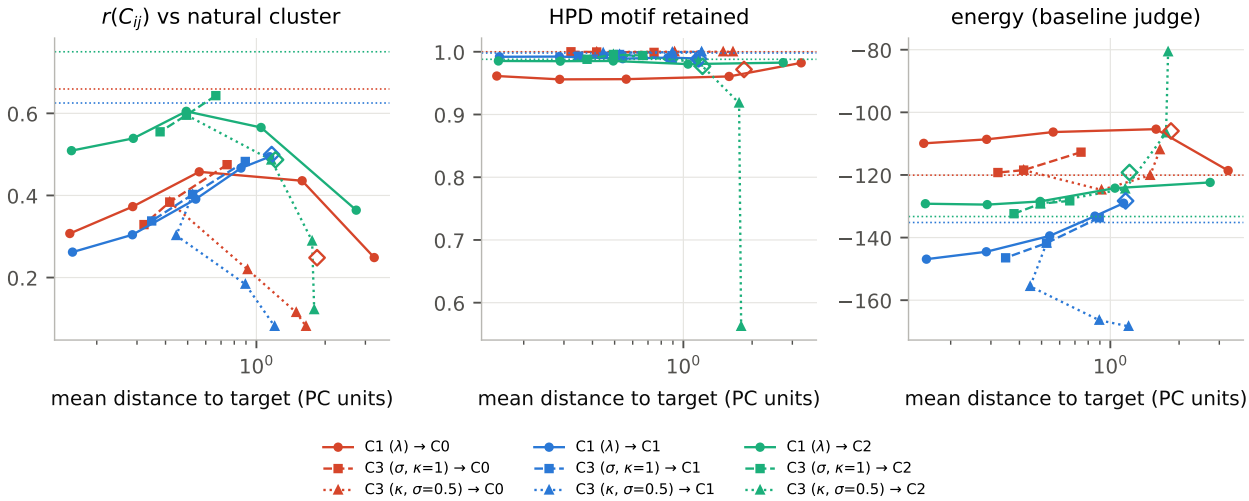


Figure 3: Accuracy–quality trade-off, one curve per method and target (colour). Solid circles: C1, $\lambda = 0.25 \to 64$ (right to left). Dashed squares: C3 at $\kappa = 1$, $\sigma = 1 \to 0.25$. Dotted triangles: C3, $\sigma = 0.5$, $\kappa = 1 \to 8$. Open diamonds: rejection from the unconditional baseline. Dotted horizontal lines: natural ceiling ($r(C_{ij})$) and natural cluster value (HPD, energy).

A Per-target tables and convergence

Columns: mean distance to target; overshoot (signed, > 0 beyond the target); % landing in the target cluster; $r(f_i)$, $r(C_{ij})$ against the natural test sequences of the cluster; % with the HPD motif; mean energy under the baseline judge; mean distance to the closest training sequence; mean pairwise distance in the pool. All pools of size 3000 except the rejection pool.

	dist.	overshoot	in C_k (%)	$r(f_i)$	$r(C_{ij})$	HPD (%)	E	d_{train}	div.
natural (train, ceiling)	0.99	-0.75	100	0.994	0.659	100	-123.3	0.00	0.67
rejection from baseline (n=496)	1.84	-1.70	100	0.928	0.248	97	-106.0	0.49	0.69
unconditional baseline	3.89	-3.47	5	0.684	0.158	99	-126.3	0.39	0.62
C1 λ=1	1.59	-1.42	80	0.963	0.436	96	-105.4	0.48	0.70
C1 λ=4	0.56	-0.41	100	0.982	0.458	96	-106.3	0.46	0.64
C1 λ=16	0.29	-0.13	100	0.969	0.373	96	-108.6	0.44	0.61
C1 λ=64	0.15	-0.04	100	0.964	0.307	96	-109.9	0.44	0.60
C3 σ=1	0.74	-0.56	100	0.983	0.475	100	-112.7	0.45	0.63
C3 σ=0.5	0.42	-0.14	100	0.968	0.384	100	-118.4	0.42	0.57
C3 σ=0.25	0.32	-0.04	100	0.965	0.329	100	-119.2	0.41	0.56
C3 σ=0.5, κ=2	0.91	+0.84	100	0.868	0.219	100	-124.8	0.38	0.45
C3 σ=0.5, κ=4	1.49	+1.36	100	0.726	0.115	100	-120.1	0.39	0.29
C3 σ=0.5, κ=8	1.65	+1.47	100	0.635	0.081	100	-111.9	0.40	0.16

Table 2: Target C0 (the under-sampled cluster). Natural test energy -120.1.

	dist.	overshoot	in C_k (%)	$r(f_i)$	$r(C_{ij})$	HPD (%)	E	d_{train}	div.
natural (train, ceiling)	1.30	-1.16	100	0.996	0.625	100	-136.0	0.00	0.60
rejection from baseline (n=3000)	1.17	-1.02	100	0.995	0.500	99	-128.3	0.38	0.59
unconditional baseline	1.52	-1.36	86	0.992	0.425	99	-126.2	0.39	0.62
C1 λ=1	0.86	-0.72	100	0.993	0.467	99	-133.1	0.36	0.56
C1 λ=4	0.54	-0.40	100	0.986	0.391	99	-139.5	0.33	0.52
C1 λ=16	0.29	-0.15	100	0.978	0.304	99	-144.5	0.30	0.48
C1 λ=64	0.16	-0.05	100	0.975	0.262	99	-146.9	0.29	0.47
C3 σ=1	0.89	-0.76	100	0.995	0.483	99	-133.7	0.35	0.56
C3 σ=0.5	0.53	-0.36	100	0.987	0.402	100	-141.8	0.32	0.52
C3 σ=0.25	0.35	-0.14	100	0.977	0.338	99	-146.5	0.30	0.48
C3 σ=0.5, κ=2	0.45	+0.28	100	0.960	0.302	100	-155.6	0.26	0.42
C3 σ=0.5, κ=4	0.89	+0.84	100	0.922	0.184	100	-166.4	0.20	0.31
C3 σ=0.5, κ=8	1.20	+1.17	100	0.866	0.081	100	-168.3	0.17	0.21

Table 3: Target C1. Natural test energy -135.2.

	dist.	overshoot	in C_k (%)	$r(f_i)$	$r(C_{ij})$	HPD (%)	E	d_{train}	div.
natural (train, ceiling)	0.83	-0.08	100	0.997	0.750	99	-137.3	0.00	0.64
rejection from baseline (n=948)	1.21	-0.74	100	0.977	0.487	98	-119.2	0.45	0.68
unconditional baseline	3.29	-3.07	10	0.729	0.336	99	-127.2	0.39	0.62
C1 λ=1	1.05	-0.55	88	0.987	0.566	98	-124.2	0.44	0.65
C1 λ=4	0.49	+0.00	100	0.993	0.605	99	-128.5	0.43	0.62
C1 λ=16	0.29	+0.03	100	0.992	0.539	98	-129.5	0.43	0.61
C1 λ=64	0.15	+0.01	100	0.990	0.509	99	-129.2	0.43	0.61
C3 σ=1	0.66	+0.03	100	0.995	0.643	99	-128.3	0.43	0.62
C3 σ=0.5	0.49	+0.11	100	0.994	0.596	100	-129.2	0.42	0.61
C3 σ=0.25	0.38	+0.13	100	0.991	0.555	99	-132.4	0.42	0.60
C3 σ=0.5, κ=2	1.16	+0.87	100	0.906	0.486	98	-124.5	0.43	0.59
C3 σ=0.5, κ=4	1.75	+1.11	100	0.697	0.289	92	-106.5	0.48	0.55
C3 σ=0.5, κ=8	1.79	+1.03	100	0.436	0.122	56	-80.7	0.58	0.40

Table 4: Target C2. Natural test energy -133.3.

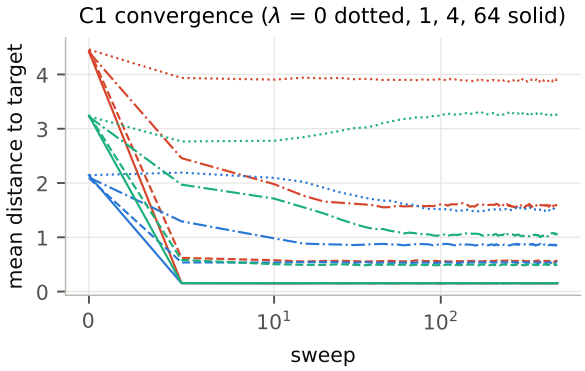


Figure 4: C1: mean distance to the target along the chains (colour = target, line style = λ).